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Where predictive information resides in endogenous A-to-I RNA editing

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A-to-I RNA editing helps prevent cellular double-stranded RNA from triggering antiviral defenses. Its selectivity remains only partly explained: neighboring adenosines can be edited differently, and the same site can differ across tissues. Existing predictors recognize many of these patterns, but their overall accuracy does not distinguish contributions from the whole RNA duplex, each target's neighborhood, or the cellular setting. We asked what each scale adds beyond a common sequence–structure reference in human duplexes formed by inverted Alu repeats. Controlled comparisons tested additional biological measurements alongside changes in representation, supervision, and model sophistication. Genomic organization added information about how extensively duplexes were edited. A correctly centered local representation improved site prediction using sequence and pairing information already present in the reference. Editing differed across tissues at matched sites, but the tested bulk contextual measurements did not consistently improve prediction. Accounting for dependence among sites also helped; most generic model extensions yielded limited or adverse returns under the tested regime. A frozen two-view predictor improved performance across all five held-out benchmark endpoints. The comparisons distinguish gains obtainable from existing RNA descriptions from contextual variation left unresolved by available measurements, motivating tests with editing and its candidate regulators measured in the same cells and transcripts.

Significance

RNA editing helps prevent a cell's own RNA from triggering antiviral defenses. Neighboring adenosines can be edited differently, and the same site can differ across tissues. Existing predictors recognize many of these patterns, but prediction accuracy alone does not explain what information distinguishes whole RNA duplexes, individual sites, and tissue settings. We compared the contributions of these information sources. Genomic organization and focusing models on individual adenosines improved different aspects of prediction; tested bulk measurements did not consistently resolve tissue-associated variation. Most generic model extensions also offered limited returns. These comparisons identify useful information in current RNA descriptions and contextual questions that need measurements matched to editing in the same cells and transcripts.

A-to-I RNA editing | ADAR | inverted Alu | RNA structure | machine learning

Adenosine deaminases acting on RNA (ADARs) convert adenosine to inosine in double-stranded RNA (dsRNA). This editing helps prevent a cell's own RNA from triggering antiviral defenses and can alter transcript function (1–5). Most detected human editing occurs in Alu repeats, abundant repeated sequences whose oppositely oriented copies can fold together within a transcript (6–10). These duplexes provide natural ADAR substrates, but they are not edited uniformly. Neighboring adenosines can have sharply different outcomes, and the same genomic site can differ among tissues (11, 12). Explaining this selectivity requires understanding both the RNA that can be edited and the conditions in which editing occurs.

Predicting likely editing sites is already possible. Sequence-based approaches, including neural networks and fine-tuned language models, recover substantial predictive signal (13–15). AdarEdit extended this approach to a graph of the complete RNA duplex, with nucleotides connected along the sequence and across predicted base pairs (16). These studies also examined motifs and structural sensitivity. Their success leaves unresolved what additional information predicts editing differences between duplexes, among sites within a duplex, and across tissues. An overall prediction score cannot separate these contributions, or tell us whether improvement requires new biological measurements or better use of existing inputs. That distinction matters for deciding what to measure next to understand editing selectivity.

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Biochemical and systematic perturbation studies have already identified sequence preferences and structural determinants of editing (17–21). For endogenous RNA, their contributions can be examined at three scales. Whole-duplex sequence, folding, and genomic organization describe a potential substrate (22). Transcript processing, localization, regulator exposure, and enzyme state may determine whether that substrate is available for editing in a cell. Within a duplex, the local sequence and pairing around an adenosine distinguish it from neighboring candidates. Information useful for predicting how extensively a duplex is edited need not identify which of its sites are edited. Likewise, a static RNA description need not resolve differences between tissues.

We used the compact AdarEdit sequence–structure model as a reference for testing these distinctions. Adding genomic or cellular measurements asks whether they improve prediction beyond that reference. Reorganizing the same sequence and predicted pairing asks whether the model leaves useful information unused. Sites from one duplex also share sequence, structure, and context; changing their sampling weights tests a statistical source of improvement without adding biology. Pair-disjoint comparisons and matched controls separate these questions where the retained experiments permit. Predictive increments measure recoverability under the tested data and representations, not molecular causality; unsuccessful additions constrain their tested formulation.

A parallel search examined whether richer supervision, cross-context sharing, pretrained representations, or greater capacity and integration could recover further signal without new biological measurements (Fig. 1). The program was adaptive: results determined which branches advanced, stopped, or required unavailable data. Retaining limited and unsuccessful branches made it possible to compare plausible routes to improvement, rather than report only the selected features. The aim was to determine what current inputs can recover at each biological scale and which contextual questions remain unresolved. Only the frozen synthesis entered the final test; the Methods and SI Appendix, Table S10 distinguish the evidence supporting each branch.

Results

Search design separates discovery from held-out evaluation. The comparisons separate additions to biological input from changes in how that input is learned or represented (Fig. 1). SI Appendix, Tables S2 and S10 record the comparator, controls, effects, and disposition of each branch. Donor-resolved state and in vivo conformation lacked compatible measurements and were deferred, not counted as negative results.

We analyzed 884 graph-buildable inverted-Alu substrates across four individual GTEx tissues and Combined, a pooled endpoint. Complete pairs were assigned without overlap to 566 training, 139 validation, and 179 test pairs. Discovery combined the training and validation pairs in pair-disjoint out-of-fold (OOF) comparisons with pair-clustered uncertainty. For final fitting, the published roles were restored: training fit the selected components, validation fixed checkpoints and thresholds, and the frozen specification was evaluated on the canonical test (Fig. 2). Development comparisons support branch-level interpretation; the held-out test evaluates only the final synthesis.

Whole-duplex sequence and structure define a strong but non-universal reference. The inherited whole-duplex reference already ranks editing sites well, providing a starting point for tests of additional information (16). It achieved within-context AUROC of 0.869–0.933 across the five human endpoints and 0.823–0.865 across three nonhuman datasets (Fig. 3A). These descriptive ranges span heterogeneous datasets and do not estimate a common effect.

Transfer was strongest among human endpoints. It varied more among nonhuman datasets, and human–nonhuman transfer was lower and asymmetric (Fig. 3B and C). Sequence–structure regularities extend across contexts without being universal. These datasets cannot separate the effects of lineage, substrate composition, assay, labeling, and cellular environment.

The reference can rank sites while leaving substrate availability, tissue state, and unmeasured conformations unresolved. Its AUROC is not a fraction of explained biological variance.

Genomic organization adds between-substrate information. Distance between oppositely oriented Alus is already associated with editability (22). The unresolved question was whether genomic organization adds information beyond the sequence and structure represented in the graph. Separation between Alu arms correlated negatively with edited-site fraction in every individual tissue (Spearman $\rho = -0.334$ to -0.482; Fig. 4B). This distance is absent from the site-level input, which joins the arms with a fixed separator. Low-capacity models combining geometry with graph summaries predicted pair-level edited fraction with $R^2 = 0.050$–0.277 and outperformed graph summaries alone in every tissue (Fig. 4C; SI Appendix). Genomic organization therefore contributes reproducible between-substrate signal beyond the tested graph summaries. The modeled outcome is editing propensity; duplex realization and ADAR engagement were not measured.

Editing differs across tissues, but current bulk measurements do not resolve the variation. Editing outcomes varied among tissues even when genomic site identity was fixed. In a post-hoc analysis of development pairs, 1,221 of 74,758 sites observed in at least two individual tissues switched strictly between edited ($\geq 15\%$) and nonedited ($< 1\%$) states. Among 50,162 sites observed in all four tissues, 17.4% spanned at least 15 EditingIndex points. Alu-pair rankings on identical site sets still correlated across every tissue pair ($\rho = 0.727$–0.921 for mean EditingIndex; SI Appendix, Fig. S7). Stable substrate differences coexist with tissue-dependent site outcomes.

Predicted substrate burden did not translate reliably into better site-level classification. An outer-isolated gate added 0.0084 AUPRC and 0.0041 F1 over expanded row-wise training, but the F1 interval included zero and Brier score increased by 0.0051. A neural burden auxiliary tested against the stronger pair-first model also failed its prespecified criterion. Static geometry predicts pair-level propensity without identifying why individual sites depart from it.

Known RNA-binding protein (RBP) regulation makes occupancy and abundance plausible sources of information

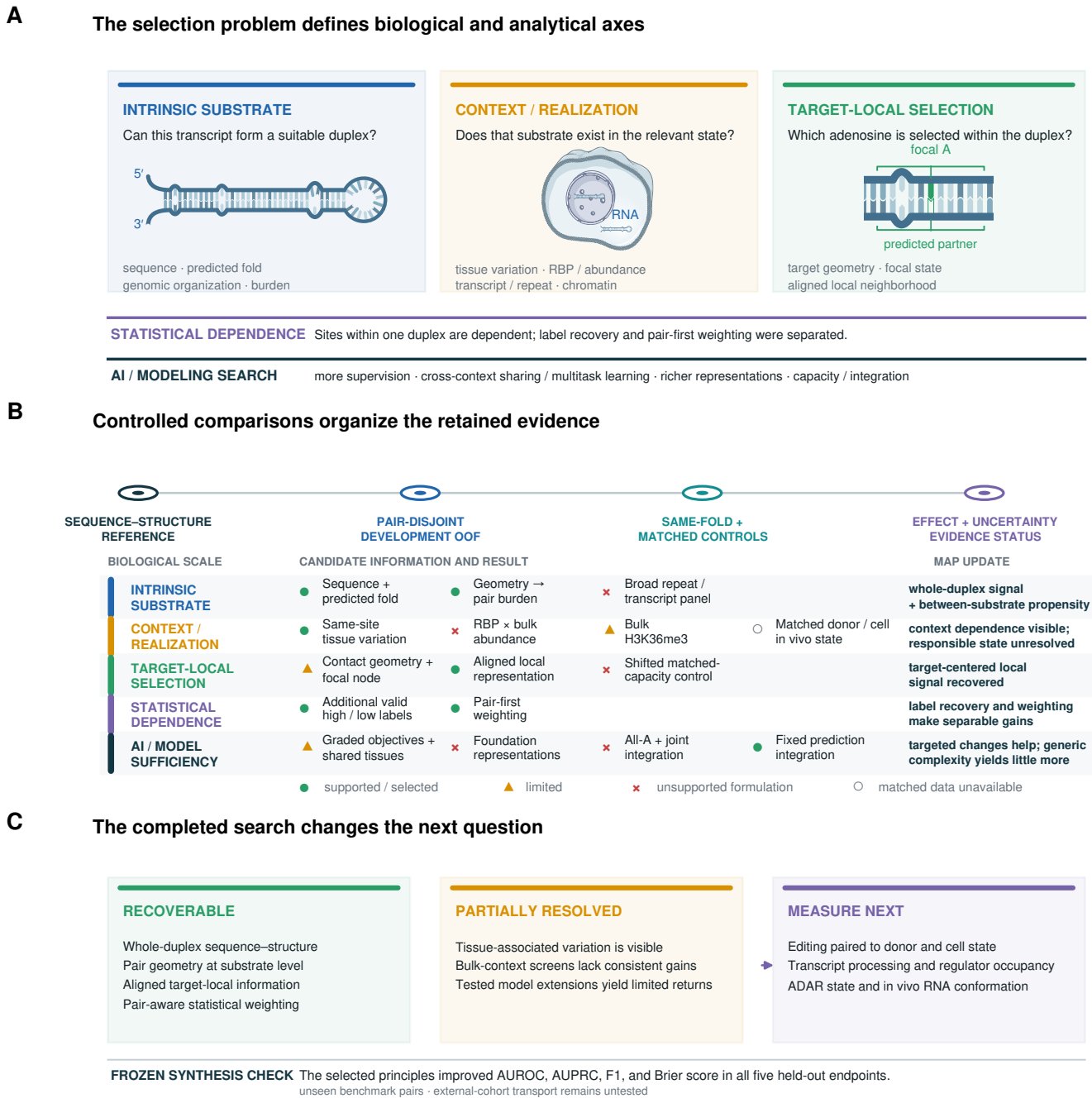


Fig. 1. Searching for information beyond an already predictive RNA representation. The comparisons ask whether further gains require added biological measurements or better use of existing inputs. (A) Whole-duplex suitability, transcript/cellular context, and local adenosine selection define the biological questions; dependent observations and model sufficiency are separate analytical questions. Illustrations are conceptual. (B) Candidates were compared against a sequence–structure reference using pair-grouped development evaluation and relevant matched controls. Symbols distinguish supported/selected, limited, unsupported in the tested formulation, and deferred branches. Early residual screens were exploratory, not fully nested (Methods). (C) Recoverable signal and unresolved context motivate finer matched measurements. The frozen test evaluated only the selected synthesis on unseen benchmark pairs. SI Appendix, Table S10 records branch-level controls, evidence regimes, and dispositions.

beyond the RNA substrate (23–26). Screens of replicated ENCODE eCLIP occupancy, sequence-based RBP predictions, tissue-average abundance and their interactions yielded no consistent residual site-level increment beyond the reference and randomized or shuffled controls (27–30). Sequence-based predictions complemented eCLIP, for which Alu multimapping makes peak absence ambiguous. These early residual screens reused pooled first-stage OOF scores and are exploratory rather than fully nested (Methods). Separate pair-disjoint analyses of matched-site switches and continuous tissue differences also found no increment from the RBP proxies beyond tissue effects. The negative evidence concerns the tested proxies, not the biological role of these regulators.

Transcript and repeat annotations likewise lacked a stable increment beyond the reference. Bulk H3K36me3 overlap yielded metric-specific effects, including a small liver AUPRC increment, without consistent site- or pair-

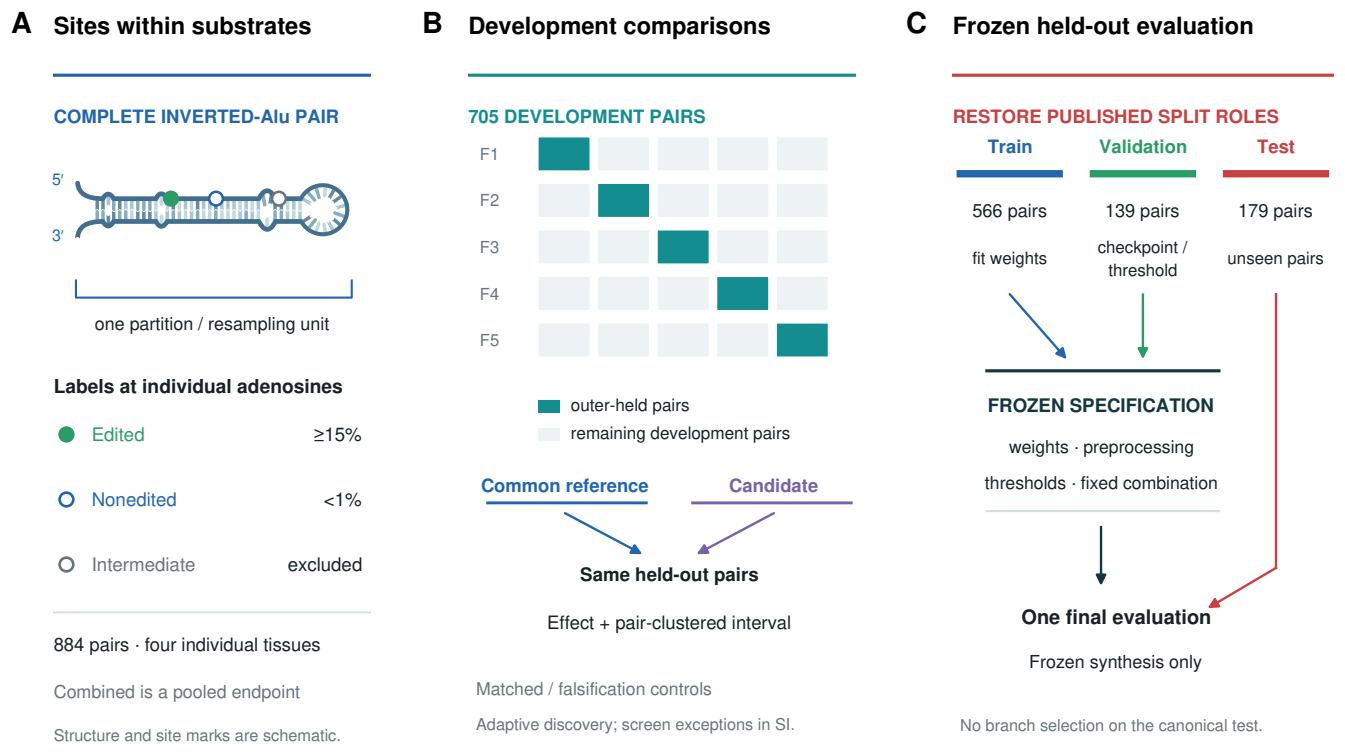


Fig. 2. Benchmark units and evaluation regimes. (A) Site labels are nested within 884 complete inverted-Alu pairs; structure and marked sites are schematic. Intermediate observations are excluded from the canonical task; Combined is pooled. (B) Five pair-disjoint outer folds define development comparisons. Neural candidate and reference models use the same folds and pair-clustered uncertainty; early pooled-score residual screens have the nestedness limitation described in Methods. Matched or falsification controls test specific attributions. (C) Final fitting restores the published split roles and freezes the specification before evaluating 179 unseen pairs. This test evaluates the synthesis, not each adaptively examined branch.

level benefit (31, 32). Thus the available contextual measurements have not supplied a reliable site-level correction in the tested comparisons. They do not identify the source of the remaining error.

Pair-first weighting balances dependent observations across substrates. Prediction can improve without a new biological measurement if the available observations are used differently. The canonical row set contains only part of the available high- and low-editing observations. Under uniform site sampling, a substrate's influence grows with its number of observations even though sites in one duplex share sequence, structure, genomic context, and often editing behavior.

A same-fold ablation separated label recovery from pair weighting in muscle (Fig. 5A; SI Appendix). Adding all high/low observations under row-wise sampling improved AUROC by +2.04 percentage points (95% pair-clustered interval +1.06 to +3.00) and F1 by +1.42 points (+0.21 to +2.57). Holding that pool and class balance fixed, then sampling a pair uniformly within each class before a site, contributed a further +2.27 AUROC points (+1.25 to +3.36) and +2.85 F1 points (+1.68 to +4.01). Label recovery and within-class substrate weighting made separable contributions.

The pair-first formulation improved the same-fold reference in all five endpoints (Fig. 5B); all 15 AUROC, AUPRC, and F1 intervals excluded zero. Pair-first sampling is therefore a weighting correction for dependent observations in this benchmark, not evidence for an additional biological determinant.

Pair-first learning improved log loss most for pairs with the fewest eligible sites, whereas the two largest site-count quintiles favored row-wise learning (SI Appendix, Fig. S8B).

Target-centered local context adds information beyond whole-duplex pooling. The reference graph already marks the target adenosine and contains predicted pairing edges. The question was whether its whole-duplex pooling used that local information effectively. Exploratory target-contact features improved ranking and Brier score but not F1. In a controlled neural comparison, explicitly retaining the focal graph node improved AUROC and AUPRC beyond an otherwise matched duplicate-global control. This localized signal unused by the pooled representation near the target.

The selected local-augmented graph predictor uses a 65-position representation centered jointly on the focal adenosine and its dot-bracket-aligned helix partner. A low-capacity dilated convolution encodes nucleotide identity, pairing state, focal and partner indicators, and signed relative position. Its output is fused with the whole-duplex graph embedding before classification. In muscle OOF data, the model reached AUROC/AUPRC/F1 of 0.9045/0.8936/0.8337, compared with 0.8885/0.8730/0.8208 for the pair-first whole-duplex-only predictor (Fig. 5C).

A matched-capacity control retained the same graph-plus-convolution architecture and optimization but shifted the local center by 96 nt. Performance fell to 0.7122/0.6943/0.7028. Aligned-minus-shifted differences

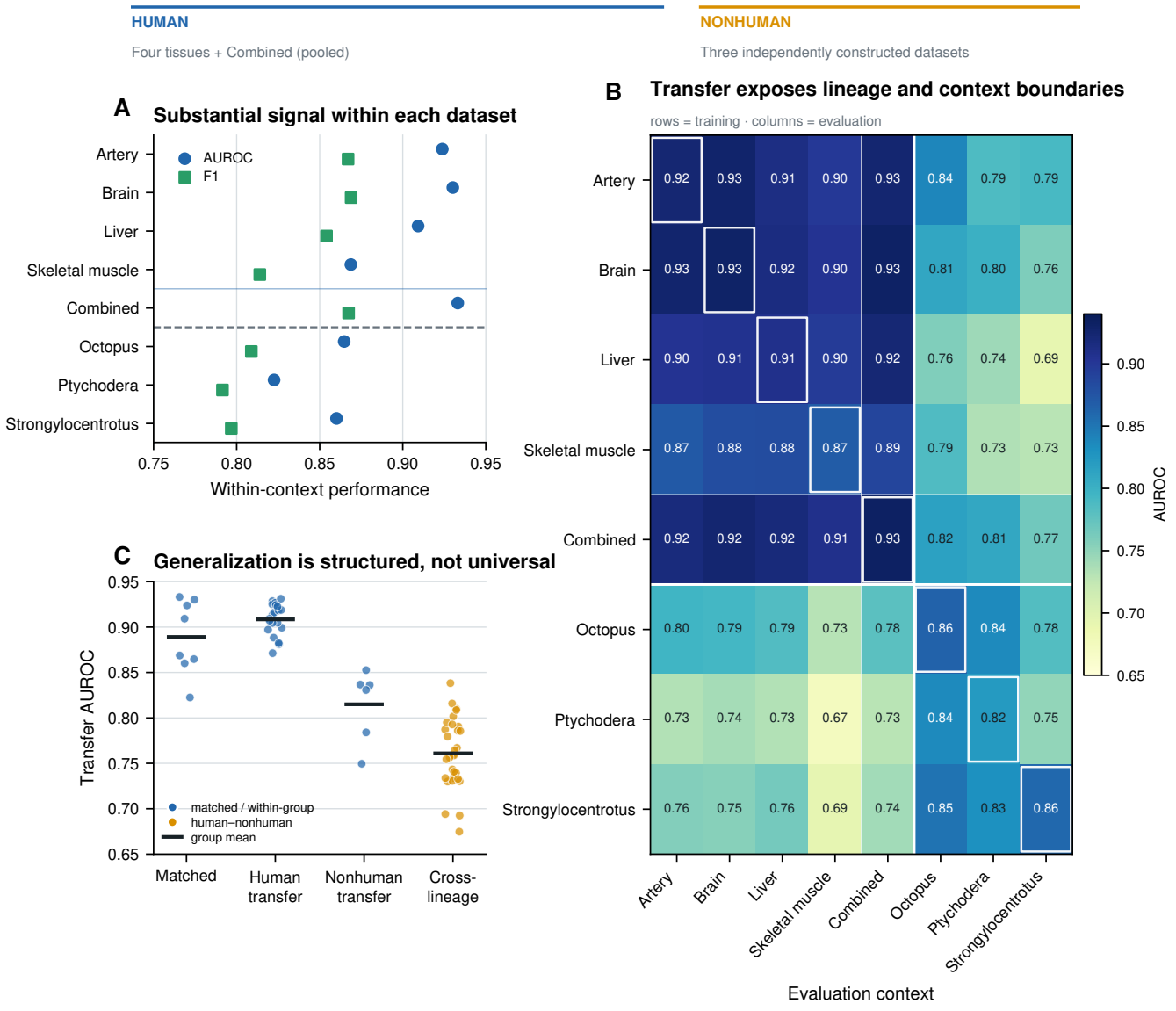


Fig. 3. Sequence and predicted structure carry substantial but non-universal signal. (A) Within-context AUROC and F1 for the retained compact sequence–structure baseline across five human endpoints and three nonhuman datasets. Combined is pooled. (B) AUROC when each training context is evaluated in every context; white lines separate human and nonhuman blocks, and outlined diagonal cells mark matched-context evaluations. (C) Transfer grouped by the relation between training and evaluation contexts; horizontal bars are group means. Dataset heterogeneity precludes causal attribution or meta-analysis.

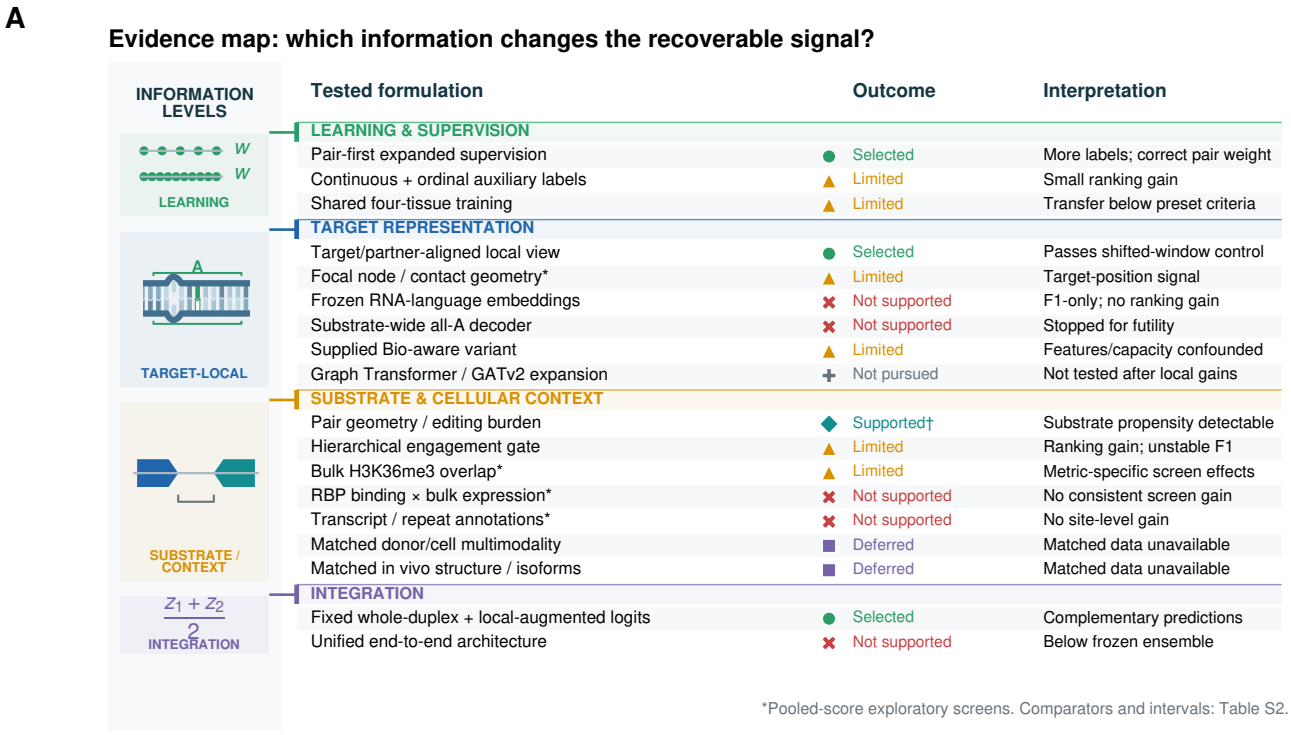
were 0.1926 AUROC, 0.1993 AUPRC, and 0.1311 F1, with entirely positive pair-clustered intervals. The correctly centered, pairing-organized neighborhood is therefore important beyond added convolutional capacity. Shifting the full window does not isolate the predicted partner from that neighborhood. The aligned model's advantage over whole-duplex prediction also held in liver and in the secondary natural-prevalence analysis; the shifted-window control was tested in muscle.

Descriptive Brier improvements occurred for paired and unpaired targets and across every target-stem extent (SI Appendix, Fig. S8C and D). Gains were not confined to one of these structural strata.

Whole-duplex and target-local views remain complementary. The whole-duplex and local-augmented predictors made sufficiently different errors to benefit from combination. A fixed mean of logits improved the whole-duplex-only predictor by 0.0256/0.0308/0.0139 AUROC/AUPRC/F1 in muscle and 0.0182/0.0214/0.0179 in liver. All six pair-clustered intervals excluded zero (Fig. 5D).

The final predictor uses two independently trained pair-first components: a whole-duplex-only graph predictor and a graph predictor augmented with the aligned target/helix-partner convolution. Their logits are combined by a fixed 0.5/0.5 mean, with no weight search (Fig. 5E). This fixed combination retains both predictive views without fitting a flexible stacker to only hundreds of independent substrates.

In a post-hoc equal-pair analysis restricted to duplexes containing both labels, the fixed ensemble improved mean within-duplex AUROC over the pair-first component in muscle (+1.78 points; 95% pair-bootstrap interval +1.21 to +2.40) and liver (+1.38 points; +1.06 to +1.70). The



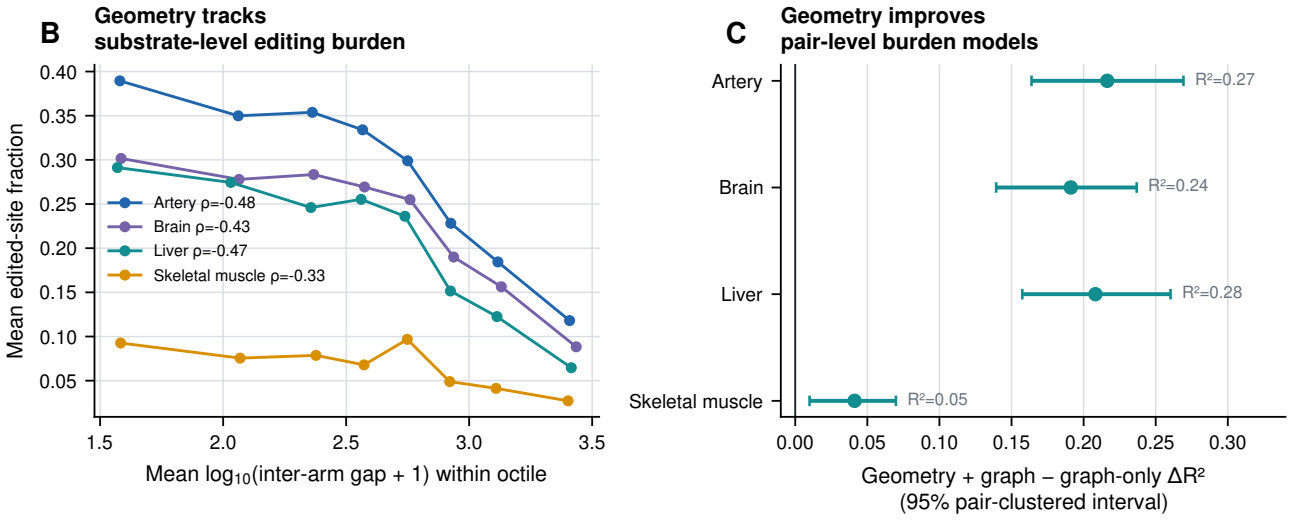


Fig. 4. The evidence map identifies a distinct substrate-level signal. (A) Search outcomes grouped by information scale. SI Appendix, Table S2 gives comparators, effects, and intervals. "Supported†" denotes an observation not yet useful for site prediction; "not supported" concerns the tested formulation. Asterisks mark pooled-score exploratory screens (Methods); within the focal/contact row, only the contact-feature screen has this limitation. Deferred measurements were unavailable. (B) Mean edited-site fraction across inter-arm-gap octiles in four tissues; ρ is calculated over individual development pairs. (C) Increment in pair-level R^2 after adding geometry to graph summaries. Bars are 95% pair-clustered intervals; labels give combined-model R^2.

aligned-local component alone showed a clear within-duplex increment only in muscle, so this analysis supports complementary within-substrate ranking rather than a replicated local-component effect (SI Appendix, Fig. S8A).

Most tested model-expansion routes yield limited returns. The remaining error could reflect limitations of learning as well as missing measurements. Four classes of model-sufficiency tests examined that possibility: richer supervision, cross-context or multitask learning, richer or pretrained representations, and greater capacity or integration. Comparisons within each class used a compatible reference. Expanding the pool of valid high/low labels helped, whereas binary–ordinal–continuous supervision added only a small ranking effect and reduced natural-prevalence performance when combined. Shared four-

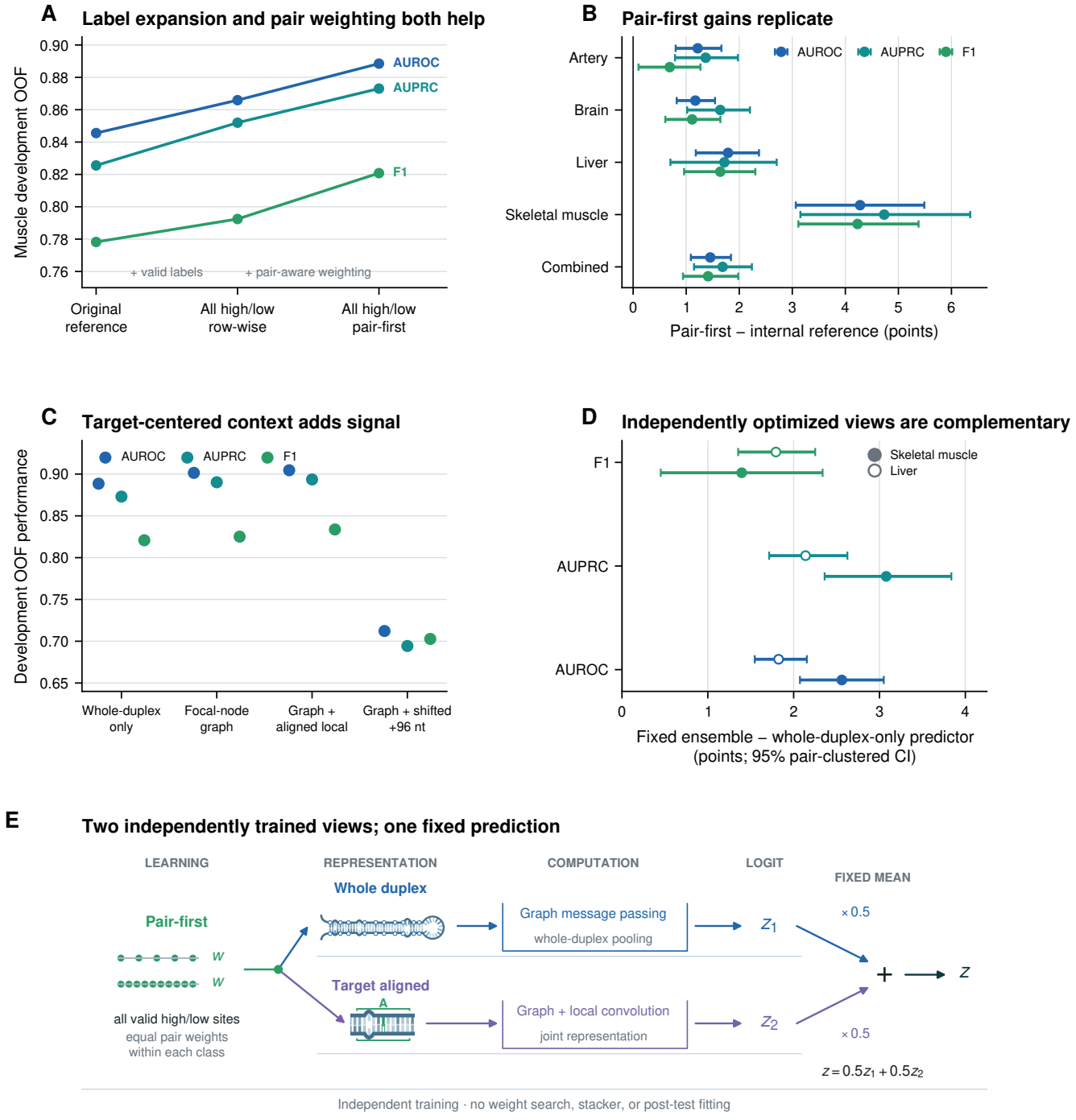


Fig. 5. Pair-first learning and target-centered representation provide complementary gains. (A) Same-fold ablations separate recovery of additional valid high/low labels from pair-first weighting. (B) Pair-first expanded supervision improves the internal reference across all five endpoints; points and bars are pair-clustered medians and 95% intervals. (C) Whole-duplex pooling, focal-state retention, and aligned-local fusion form an evidence ladder; an otherwise identical local model is shifted by 96 nt. (D) A fixed equal-logit ensemble improves its whole-duplex-only component in muscle and liver. (E) The predictor combines independently trained whole-duplex-only and graph-plus-local models without searched weights or post-test fitting. The RNA illustrations are conceptual. All panels show development evidence.

tissue learning produced limited transfer and worse mean Brier score (Fig. 4A; SI Appendix).

Representation changes were selective. Focal-node retention and aligned local encoding improved target prediction; frozen RiNALMo embeddings did not add ranking signal beyond matched controls (33). Substrate-wide decoding and the preregistered unified end-to-end model underperformed their references, while burden-based hierarchical formulations remained limited. The supplied Bio-aware variant changed both features and architecture, so it did not isolate capacity; a separate graph-transformer expansion was not pursued. Among completed controlled tests, substrate-aware weighting and target-centered representation supplied clearer gains than the tested generic extensions. This comparison does not define a ceiling for future architectures.

A frozen synthesis improves prediction on unseen benchmark pairs. The final test asked whether the selected learning and representation changes improved prediction together on unseen benchmark pairs. All components, checkpoints, thresholds, weights, code, and hashes were frozen before canonical held-out evaluation. The selected predictor improved AUROC, AUPRC, and F1 in all five reported endpoints (Fig. 6A), with mean gains of 0.0277, 0.0261, and 0.0253 over reproduced supplied-checkpoint AdarEdit. Every endpoint had a positive 95% pair-clustered interval for all three metrics, and Brier score decreased throughout (Fig. 6B and C; SI Appendix, Table S1).

Combined overlaps the individual tissue endpoints, so the mean is descriptive rather than a five-tissue meta-analysis. The frozen combination generalizes to unseen pairs within this benchmark. Its component-level attribution rests on the controlled development comparisons above.

At frozen validation-selected thresholds, the selected predictor corrected 32.0–50.6% of baseline false positives while converting 1.7–5.9% of baseline true positives into false negatives (Fig. 6D). A secondary matched-recall analysis is reported in SI Appendix, Table S9.

Discussion

The additional information about editing did not all require new measurements. Genomic organization supplied information absent from the folded RNA input, whereas target-centered encoding recovered site-level signal from sequence and pairing already present. The two gains resolve different parts of the selection problem: how extensively a substrate is edited and how well individual adenosines can be distinguished. Accounting for dependent sites supplied a separate statistical improvement. These distinctions explain why a better score alone would have been an incomplete answer: it would not identify whether the improvement came from adding biology, preserving local information, or changing the influence of correlated observations.

Earlier work connected Alu geometry with aggregate editing and used biochemical assays, systematic perturbations, and predictive models to identify local sequence and structural preferences (17–22). The present comparisons ask what those sources contribute beyond an already predictive RNA representation. Geometry explained pair-level burden beyond graph summaries, but did not become a reliable site-level correction. Conversely, the aligned-versus-shifted control recovered additional site-level signal without introducing a new measurement of RNA structure. Correctly centered, pairing-organized context matters beyond whole-duplex pooling; the experiment does not isolate the predicted partner or establish a molecular mechanism.

The context branch identifies a measurement hypothesis, not an observed latent state. Perturbation studies implicate RBPs and chromatin-associated regulation in editing (23–26, 31). Here, tissue-associated differences coexist with limited predictive value from cell-line occupancy, tissue-average abundance, transcript and repeat annotations, and bulk chromatin. These findings motivate measuring regulator exposure and transcript state at the same biological resolution as editing. They do not establish that coarse measurement is the sole reason the tested proxies underperformed.

The modeling search tested how much of the remaining predictive gap could be reduced without adding biological measurements. Graded supervision and cross-tissue sharing gave limited gains; frozen RNA foundation representations, substrate-wide decoding, and connected end-to-end integration did not outperform their respective references on the selection criteria. Class-balanced pair-first learning and correctly centered local encoding supplied clearer gains, and their frozen combination improved held-out discrimination, classification, and probability accuracy. Generic model sophistication alone did not close the gap in the tested regime. Residual error could still reflect finite data, model misspecification, measurement noise, structural uncertainty, unmeasured transcript or cellular state, or biological stochasticity. The present data cannot apportion these sources.

The benchmark covers graph-buildable human inverted-Alu substrates and tissue summaries. Predicted structure is a surrogate for in vivo conformations; eCLIP is cell-line-specific and repeat-censored. Nonhuman transfer confounds lineage, substrate class, assay, and labeling. The canonical labels are balanced and exclude intermediate observations, so natural-prevalence analyses remain secondary. Component attribution comes from adaptively selected, nonmultiplicity-adjusted development comparisons; early pooled-score residual screens additionally require fully nested replication. Tissue comparisons share substrates and are not independent cohorts. The frozen test assesses unseen pairs within the benchmark construction, not transportability to an independently generated human cohort (16).

The next biological test is to measure editing and its candidate regulators on the same RNA substrates in the same donors or cells. Transcript processing, ADAR isoform and interferon state, localization, regulator occupancy, quantitative chromatin, and in vivo RNA conformation are motivated candidates, not a proven sufficient measurement set. Their value should be judged by whether they improve prediction of editing differences at the same sites across biological contexts, beyond what sequence, predicted pairing, and genomic organization already predict.

Materials and Methods

Search design and evidence classification. The adaptive search grouped candidates by intrinsic substrate properties, transcript/cellular context, and target-local selection, with separate axes for dependent observations and model sufficiency. The latter covered richer supervision, cross-context sharing, richer/pretrained representations, and capacity or integration. Priorities reflected biological rationale, information beyond the reference, measurement compatibility, leakage risk, and effective sample size. Development branches used pair-grouped comparisons against a compatible reference, with randomized, shifted-location, or matched-capacity controls where relevant. Selected branches met recorded promotion criteria and entered the synthesis; supported findings were retained without selection; limited effects were metric- or context-specific. Not supported denotes failure of the tested formulation. Deferred branches lacked suitable data or a validated resource; superseded branches were not pursued after stronger evidence emerged. These adaptive comparisons are not multiplicity-adjusted confirmation, and the coverage taxonomy is not a claim that the entire program was preregistered. SI Appendix, Tables S2 and S10 distinguish controlled comparisons, exploratory screens, descriptive characterization, and untested branches. Only the frozen synthesis entered the final canonical evaluation.

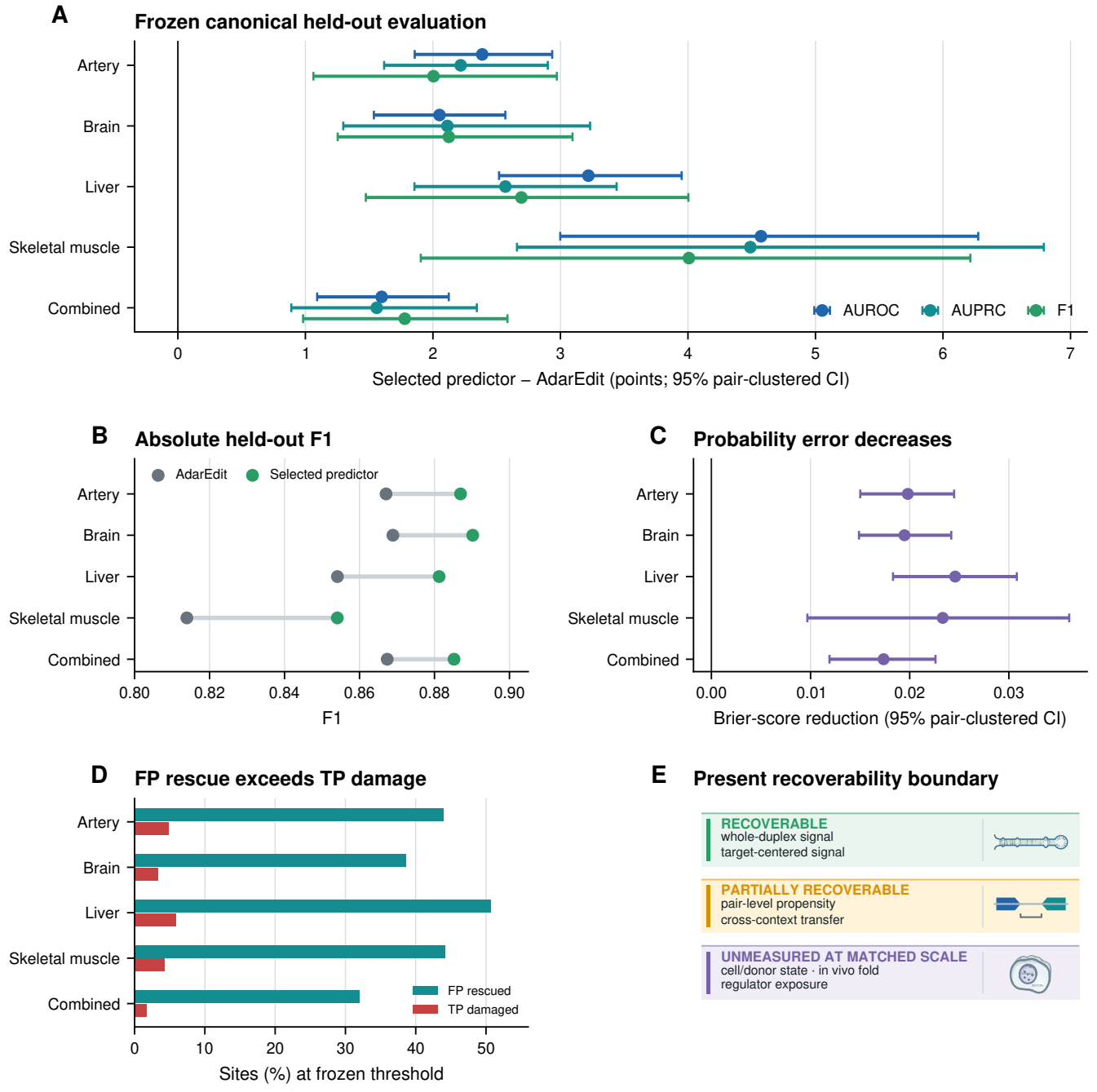


Fig. 6. Frozen evaluation confirms within-benchmark generalization and summarizes what current measurements recover. (A) Selected predictor minus reproduced supplied-checkpoint AdarEdit for AUROC, AUPRC, and F1; bars are 95% pair-clustered bootstrap intervals. (B) Absolute F1. (C) Reduction in Brier score. (D) Baseline false positives corrected and baseline true positives converted to false negatives at frozen validation-selected thresholds. (E) Information captured, partially resolved, and unavailable at matched biological scale; tier-specific icons are conceptual. Combined is pooled. All quantitative panels use the frozen canonical held-out evaluation.

Benchmark and labels. We used the AdarEdit v2 reproducibility release at upstream revision `5c8b57d70cdc2eb1c5e168431013275bcc9ed0c5`. The benchmark contains 884 graph-buildable inverted-Alu pairs. Human sites were labeled edited at EditingIndex $\geq 15\%$, nonedited at EditingIndex $< 1\%$, and intermediate otherwise. The canonical task excludes intermediate observations. We verified pair identities and the fixed 566/139/179 train/validation/test assignment before analysis. Combined was used only as its published pooled endpoint.

Development protocol. Discovery used the original train and validation pairs in 5 outer folds by pair ID. Neural comparisons excluded the outer fold from training, preprocessing, checkpoint choice, and threshold choice; one other fold provided internal validation. Early residual screens instead fitted pair-disjoint second-stage models to pooled first-stage OOF logits. Their meta-training scores could depend on the current outer fold through first-stage fitting, so they are exploratory, not fully nested estimates (SI Appendix, section A). Later outer-isolated gates and foundation fusion used fold-specific first-stage fits. Supplied checkpoints were reproduced using aggregate test metrics before discovery; candidate development did not access the canonical test.

Models. The first component retained the compact AdarEdit graph attention encoder. Each epoch allocated equal high/low class counts, sampled pairs uniformly among those eligible for each class, and then sampled a site within each selected pair. The canonical-row update budget was unchanged. The second component was trained independently with the same graph encoder. It added a 65-position target/helix-partner-aligned representation encoded by two dilated convolutional layers, then fused local

and graph embeddings. The selected probability was $\sigma[(\operatorname{logit} p_{\mathrm{whole}} + \operatorname{logit} p_{\mathrm{augmented}})/2]$, with weights fixed at 0.5/0.5 and no weight search. The shifted-window control preserved architecture and optimization while moving the local center 96 nt. SI Appendix gives the feature definitions and remaining protocols.

Statistics and final evaluation. Metrics were AUROC, AUPRC, F1, precision, recall, specificity, balanced accuracy, MCC, and Brier score. Thresholds maximized validation F1 within each run. Pair-clustered bootstraps resampled Alu pairs with all their sites: development comparisons used 1,000 or 2,000 replicates, the post-hoc within-duplex analysis used 5,000, and the final comparison used 2,000. These intervals condition on the fitted models and do not include variability from rerunning training or adaptive selection. Code, inputs, weights, epochs, thresholds, and checkpoint hashes were frozen before the selected predictor's held-out evaluation; no subsequent model selection or tuning occurred. Natural-prevalence and matched-recall analyses are secondary (SI Appendix).

AI-assisted preparation. OpenAI Codex assisted with manuscript editing, source-code review, and figure-rendering code. RNA and cell illustrations were adapted from credited DBCLS/TogoTV and Servier Medical Art SVGs (SI Appendix, Artwork credits). Earlier GPT Image 1.5 illustrations are excluded. Author confirmation of the complete tool/model-version disclosure and of journal eligibility for AI-assisted diagram code is pending.

Data, Materials, and Software Availability. Code, experiment configurations, registries, source tables, figure scripts, retained derived artifacts, frozen checkpoints, and provenance are available at https://github.com/Scientific-Computing-Lab/ADARPro. Third-party downloads and deterministic matrices omitted from GitHub are listed with source URLs, roles, checksums, and regeneration instructions in the data manifest. The AdarEdit benchmark is available from https://github.com/Scientific-Computing-Lab/AdarEdit. No new human participants were recruited and no new primary sequencing data were generated.

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